



# Powermax125®

Professional-grade plasma metal cutting and gouging system for handheld cutting 38 mm and mechanized piercing 25 mm.



| Capacity                 | Thickness | Cut speed  |
|--------------------------|-----------|------------|
| Cutting                  |           |            |
| Recommended              | 38 mm     | 457 mm/min |
|                          | 44 mm     | 250 mm/min |
| Severance (hand cutting) | 57 mm     | 125 mm/min |
| Pierce*                  | 25 mm     |            |

\* Pierce rating for handheld use or with automatic torch height control

| Capacity      | Metal removal rate | Groove profile*                   |
|---------------|--------------------|-----------------------------------|
| Gouging       |                    |                                   |
| Typical gouge | 12,52 kg/hr        | 4,3 mm–7,9 mm D x 6,1 mm–9,9 mm W |

\* Actual groove profile varies with torch angle, arc stretch, and technique

## Maximum productivity

- Finish jobs faster with cut speeds 5 times greater than oxyfuel on 12 mm mild steel.
- Spend less time on grinding and edge preparation due to superior cut and gouge quality.
- Maximize your cutting time with 100% duty cycle.

## Low operating costs

- Reduce costs with consumable life up to four times longer than other systems in this amperage range.
- Electrode end-of-life detection protects the torch and workpiece from damage by automatically stopping power when the electrode is overused.

## Easy-to-use for cutting and gouging

- No need to change the air pressure. Smart Sense™ technology ensures that it is always correctly set.
- Tackle different jobs with multiple, easy-to-use torch styles.

## Built and tested to withstand the harshest conditions

- Duramax® Hyamp™ torches are designed for high-impact and heat resistance.
- SpringStart™ technology ensures consistent starting and a more reliable torch.
- Low maintenance maximizes uptime.

Duramax Hyamp standard torch styles  
(for more torch options, see [www.hypertherm.com](http://www.hypertherm.com))



85° hand torch



15° hand torch

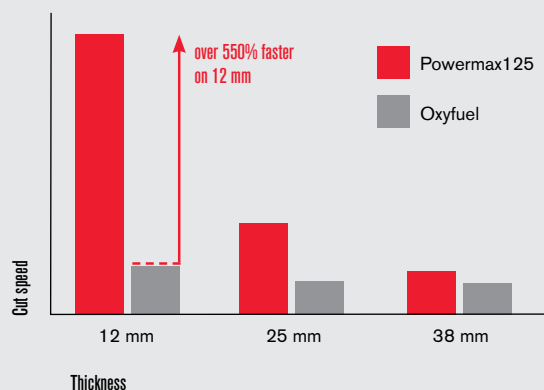


180° mini machine torch



180° full-length machine torch

Relative cut performance on mild steel



Specifications

|                                          |     |                                                                                                                                                                                  |                    |
|------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Input voltages                           | CCC | 380 V, 3-PH, 50/60 Hz                                                                                                                                                            |                    |
|                                          | CE  | 400 V, 3-PH, 50/60 Hz                                                                                                                                                            |                    |
| Kilowatt output                          |     | 21,9 kW                                                                                                                                                                          |                    |
| Input current @ 21,9 kW                  |     | 380 V, 3-PH, 38 A<br>400 V, 3-PH, 36 A                                                                                                                                           |                    |
| Output current                           |     | 30-125 A                                                                                                                                                                         |                    |
| Rated output voltage                     |     | 175 VDC                                                                                                                                                                          |                    |
| Duty cycle @ 40° C                       |     | 100% @ 125 A, 380/400 V, 3-PH                                                                                                                                                    |                    |
| Open circuit voltage (OCV)               |     | 380 V CCC<br>400 V CE                                                                                                                                                            | 290 VDC<br>305 VDC |
| Dimensions with handles                  |     | 592 mm D; 274 mm W; 508 mm H                                                                                                                                                     |                    |
| Weight w/7,6 m torch                     |     | 380 V<br>400 V                                                                                                                                                                   | 45,2 kg<br>48,9 kg |
| Gas supply                               |     | Clean, dry, oil-free air or nitrogen                                                                                                                                             |                    |
| Recommended gas inlet flow rate/pressure |     | Cutting: 260 l/min @ 5,9 bar<br>Gouging: 212 l/min @ 4,1 bar                                                                                                                     |                    |
| Power supply type                        |     | Inverter – IGBT                                                                                                                                                                  |                    |
| Engine drive requirement                 |     | 40 kW for full 125 A output                                                                                                                                                      |                    |
| Certifications                           |     | CE, C-Tick, CU/GOST, Ukr, and Serbian — for use in Europe, Australia, Belarus, Kazakhstan, Russia, Serbia, Ukraine, and other countries where accepted.<br>CCC for use in China. |                    |
| Warranty                                 |     | Power supplies have a 3-year warranty and torches a 1-year warranty.                                                                                                             |                    |



Ordering information

Below are the standard system configurations, which include a power supply, torch and work cable.

|                                                             |                        | 85° hand torch |        | 85° and 15° hand torch |        | 180° full-length machine torch |                        |                              |                     |                        |                              | 180° full-length machine torch and 85° hand torch |
|-------------------------------------------------------------|------------------------|----------------|--------|------------------------|--------|--------------------------------|------------------------|------------------------------|---------------------|------------------------|------------------------------|---------------------------------------------------|
|                                                             |                        | 7,6 m          | 15,2 m | 7,6 m                  | 15,2 m | 7,6 m                          |                        |                              | 15,2 m              |                        |                              | 15,2 m/7,6 m                                      |
|                                                             |                        |                |        | without remote pendant |        | with remote pendant            | without remote pendant | with I/O cables (no pendant) | with remote pendant | without remote pendant | with I/O cables (no pendant) | with remote pendant                               |
| Power supply with CPC port and voltage divider              | 380 V CCC <sup>1</sup> | 059556         | 059557 |                        |        | 059558                         | 059583                 |                              |                     | 059584                 |                              | 059559                                            |
|                                                             | 400 V CE               | 059526         | 059527 | 059528                 | 059572 | 059530                         |                        |                              | 059531              |                        |                              | 059529                                            |
| Power supply with CPC port, voltage divider, and erial port | 380 V CCC <sup>1</sup> |                |        |                        |        |                                | 059560                 |                              |                     | 059561                 |                              |                                                   |
|                                                             | 400 V                  |                |        |                        |        |                                | 059532                 | 059534                       |                     | 059533                 | 059535                       |                                                   |

<sup>1</sup> Does not include a power cord.



Custom configurations (select power supply, torch, consumable kit, work cable, and other components)

Power supply¹ options

|           | Power supply<br>with CPC port and voltage divider | Power supply<br>with CPC port, voltage divider, and serial port |
|-----------|---------------------------------------------------|-----------------------------------------------------------------|
| 380 V CCC | 059490                                            | 059491                                                          |
| 400 V CE  | 059486                                            | 059487                                                          |

¹ Does not include a power cord.

Starter kit consumable options

|     | Handheld | Mechanized | Mechanized with<br>ohmic retaining cap |
|-----|----------|------------|----------------------------------------|
| Kit | 428099   | 428100     | 428101                                 |

Torch options

|              | Hand torches |        | Machine torches |           | Robotic torches |        |        | Long torches |            |            |            |            |
|--------------|--------------|--------|-----------------|-----------|-----------------|--------|--------|--------------|------------|------------|------------|------------|
| Cable length | 85°          | 15°    | 180°            | Mini 180° | 45°             | 90°    | 180°   | 0,6 m, 45°   | 0,6 m, 90° | 1,2 m, 45° | 1,2 m, 90° | 1,8 m, 90° |
| 4,5 m        |              |        | 059519          | 059514    |                 |        |        |              |            |            |            |            |
| 7,6 m        | 059492       | 059495 | 059520          | 059515    | 059564          | 059565 | 059566 | 059562       | 059563     | 059567     | 059568     | 059623     |
| 10,7 m       |              |        | 059521          | 059516    |                 |        |        |              |            |            |            |            |
| 15,2 m       | 059493       | 059496 | 059522          | 059517    |                 |        |        | 059579       | 059580     | 059581     | 059582     | 059624     |
| 22,8 m       | 059494       | 059497 | 059523          |           |                 |        |        |              |            |            |            |            |
| 45,7 m       | 059664       | 059665 |                 | 059666    |                 |        |        |              |            | 059667     | 059668     |            |

\*45,7 m torch leads are compatible only with Powermax125 systems with a serial number of 125-007944 or higher.

Cable options

|              | Work cables |               |               | Control cables |                                             |                                                |                                             |                                        |                                 |
|--------------|-------------|---------------|---------------|----------------|---------------------------------------------|------------------------------------------------|---------------------------------------------|----------------------------------------|---------------------------------|
| Cable length | Hand clamp  | C-style clamp | Ring terminal | Remote pendant | CNC, spade<br>connector,<br>divided voltage | CNC, spade<br>connector, no<br>divided voltage | CNC, D-sub<br>connector,<br>divided voltage | RS-485<br>Serial comm,<br>unterminated | RS-485<br>Serial comm,<br>D-sub |
| 7,6 m        | 223292      | 223298        | 223295        | 128650         | 228350                                      | 023206                                         | 223048                                      | 223236                                 | 223239                          |
| 15,2 m       | 223293      | 223299        | 223296        | 128651         | 228351                                      | 023279                                         | 123896                                      | 223237                                 | 223240                          |
| 22,8 m       | 223294      | 223300        | 223297        | 128652         |                                             |                                                |                                             |                                        |                                 |
| 45,7 m       | 223737      |               | 223744        | 428755         |                                             |                                                |                                             |                                        |                                 |

Torch consumable parts

Nozzles and electrodes are available in various quantities. Contact your distributor for more information.

| Consumable type     | Torch type      | Amperage | Nozzle | Shield/<br>Deflector | Retaining cap     | Swirl ring       | Electrode |
|---------------------|-----------------|----------|--------|----------------------|-------------------|------------------|-----------|
| Drag-cutting        | Hand            | 45       | 420158 | 420172               | 220977            | 220997           | 220971    |
|                     |                 | 65       | 420169 | 420172               |                   |                  |           |
|                     |                 | 125      | 220975 | 420000               |                   |                  |           |
| Mechanized          | Machine         | 45       | 420158 | 420168               | 220977 or 420156² | 220997           | 220971    |
|                     |                 | 65       | 420169 | 420168               |                   |                  |           |
|                     |                 | 125/105  | 220975 | 220976               |                   |                  |           |
| FineCut®            | Hand<br>Machine | 45       | 420151 | 420152               | 220977 or 420156² | 420159<br>220997 | 220971    |
| Max removal gouging | Hand/Machine    | 30–125   | 420001 | 420112               | 220977            | 220997           | 220971    |
| Max control gouging | Hand/Machine    | 30–125   | 420001 | 420519               | 220977            | 220997           | 220971    |

² Ohmic retaining cap

## Recommended Hypertherm genuine accessories



### Hyamp™ deluxe circle cutting guide

Quick and easy setup for accurate circles up to 70 cm diameter. For use with the Powermax125®.

017053 Hyamp deluxe kit



### Air filtration kit

A ready-to-install kit with a 1-micron filter and an auto drain moisture separator protects against contaminated air.

228890 Filter plus cover for Powermax105/125

011092 Replacement filter element for Eliminator

101215 Cover only for Powermax105/125



### Leather torch sheathing

Available in 7,6 m sections, this option provides additional protection for torch leads against burn-through and abrasion.

024877 Black leather with Hypertherm logos



### System dust covers

Made from a flame-retardant vinyl, a dust cover will protect your Powermax system for years.

127360 Cover, Powermax125



### Wheel/gantry kits

Complete, pre-assembled kits for added mobility or mounting on a cutting table gantry.

229467 Powermax105/125 wheel kit

229570 Powermax105/125 gantry kit



### Hyamp™ torch adapter

The Hyamp torch adapter kit makes the 44 mm machine torches (on the Powermax125 and MAXPRO200® systems) compatible with track burners, pipe cutters, cutting tables, and legacy equipment that accept a 35 mm barrel torch.

428495 Hyamp torch adapter

## ISO 9001:2008

Environmental stewardship is one of Hypertherm's core values, and it is critical to our success and our customers' success. We are striving to reduce the environmental impact of everything we do. For more information: [www.hypertherm.com/environment](http://www.hypertherm.com/environment).

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